

GCE O-LEVEL MATHEMATICS PRACTICE PAPER 10

Comprehensive Review - Mixed Topics

Duration: 2.5 hours Total Marks: 100
Name: _____ Date: _____

SECTION A: Short Answer Questions (40 marks)

1. Solve: $2x^2 - 7x + 3 = 0$ (Algebra) [3 marks]

2. Find area of triangle with sides 5, 6, 7 (Geometry) [3 marks]

3. If $P(A) = 0.4$ and $P(B) = 0.3$, independent, find $P(A \cup B)$ (Probability) [3 marks]

4. Differentiate: $y = x^3 - 2x^2 + 5$ (Calculus) [2 marks]

5. Find equation of line through (2,3) with gradient 4 (Coordinate Geometry) [2 marks]

6. Solve: $3^x = 27$ (Exponentials) [2 marks]

7. Find magnitude of vector (1, 2, 2) (Vectors) [2 marks]

8. Sum of first 5 terms: 3, 6, 12, 24, ... (Sequences) [3 marks]

SECTION B: Long Answer Questions (60 marks)

9. A right circular cone has base radius 3 cm, height 4 cm. Find volume and surface area. (5 marks)

10. Integrate: $\int (2x + 1/x) dx$. (3 marks)

11. In triangle ABC, angle A = 45° , angle B = 60° , side a = 5 cm. Find side b using sine rule. (4 marks)

12. A function $f(x) = x^2 - 4x + 3$. Find: (a) roots, (b) vertex, (c) minimum value. (5 marks)

13. Two variables: $x + 2y = 5$ and $x^2 + y^2 = 10$. Solve simultaneously. (6 marks)

14. Exam problem combining 3 concepts: Differentiate $f(x) = (2x+1)/(x-1)$, find critical points, determine nature. (6 marks)

SOLUTIONS

SECTION A SOLUTIONS

Q1: Solve $2x^2-7x+3=0$

Using quadratic formula: $x = (7 \pm \sqrt{(49-24)})/4 = (7 \pm 5)/4$, $x = 3$ or $x = 0.5$

Q2: Area with sides 5,6,7

$s = (5+6+7)/2 = 9$, Area = $\sqrt{(9 \times 4 \times 3 \times 2)} = \sqrt{216} = 6\sqrt{6} \text{ cm}^2$

Q3: $P(A \cup B)$ independent

$P(A \cup B) = P(A) + P(B) - P(A)P(B) = 0.4 + 0.3 - 0.12 = 0.58$

Q4: Differentiate x^3-2x^2+5

$dy/dx = 3x^2 - 4x$

Q5: Line equation

$y - 3 = 4(x - 2)$, $y = 4x - 5$

Q6: Solve $3^x = 27$

$3^x = 3^3$, $x = 3$

Q7: Magnitude of (1,2,2)

$||v|| = \sqrt{(1 + 4 + 4)} = 3$

Q8: Sum of GP 3,6,12,24,...

$a=3$, $r=2$, $S_n = 3(2^n-1)/(2-1) = 3 \times 31 = 93$

SECTION B SOLUTIONS

Q9: Cone volume and surface area

$V = (1/3)\pi r^2 h = (1/3)\pi(9)(4) = 12\pi \text{ cm}^3$, Slant height = 5, $SA = \pi r^2 + \pi r l = 9\pi + 15\pi = 24\pi \text{ cm}^2$

Q10: Integrate $(2x + 1/x)$

$\int (2x + 1/x) dx = x^2 + \ln|x| + C$

Q11: Find side b using sine rule

$a/\sin A = b/\sin B$, $5/\sin 45^\circ = b/\sin 60^\circ$, $b = 5 \sin 60^\circ / \sin 45^\circ = 5(\sqrt{3}/2)/(\sqrt{2}/2) = 5\sqrt{3}/\sqrt{2} = 5\sqrt{6}/2 \approx 6.12 \text{ cm}$

Q12: Properties of $f(x)=x^2-4x+3$

(a) Roots: $x^2-4x+3=0$, $(x-1)(x-3)=0$, $x=1,3$; (b) Vertex: $x = -b/2a = 2$, $y = 4-8+3 = -1$, $(2,-1)$;
(c) Min = -1

Q13: Solve $x+2y=5$ and $x^2+y^2=10$

From first: $x=5-2y$, substitute: $(5-2y)^2+y^2=10$, $25-20y+4y^2+y^2=10$, $5y^2-20y+15=0$, $y^2-4y+3=0$,
 $y=1,3$, $x=3,-1$, Solutions: $(3,1)$, $(-1,3)$

Q14: Differentiate $(2x+1)/(x-1)$

Using quotient rule: $f'(x) = [2(x-1)-(2x+1)]/(x-1)^2 = -3/(x-1)^2$, No critical points (always negative), function always decreasing